Adaptive Spectral Thresholding via Kurtosis-Guided Squared Envelope Analysis for Load-Invariant Bearing Fault Detection

Member 1, Member 2, Member 3, and Member 4

***Abstract*—Rolling element bearing diagnostics rely on envelope spectrum analysis to extract the periodic impulse rates produced by localized surface defects. In industrial operation, however, variable mechanical load alters the shaft slip, the structural damping and the broadband noise floor, so that fixed bandpass filters and static spectral thresholds exhibit high false-alarm rates under heavy load and missed detections under light load. This paper presents a training-free pure Digital Signal Processing (DSP) framework that combines Kurtosis-guided resonance demodulation with local noise-floor adaptive thresholding in the Squared Envelope Spectrum (SES). Cascade second-order-section (SOS) zero-phase IIR filtering, discrete Hilbert transform demodulation, Hann-windowed Fast Fourier Transform (FFT) and order-statistic Median–Median Absolute Deviation (Median–MAD) Constant False Alarm Rate (CFAR) thresholding allow the detector to track load-induced noise-floor shifts and shaft slip automatically. The processing chain is mathematically interpretable, requires no training data, and is sized for fixed-point embedded implementation; it is applied here to the multi-load CWRU and MFPT bearing benchmarks.**

***Index Terms*—Bearing fault diagnosis, Squared Envelope Spectrum (SES), Hilbert transform, spectral kurtosis, IIR/FIR filter design, CFAR, Median–MAD adaptive thresholding.**

I. Introduction

ROLLING element bearings support rotating shafts in motors, pumps, turbines, gearboxes and many other machines. Their condition matters for machine availability, because localized surface damage can develop into severe mechanical failure when it is not found early. A defect on an inner race, outer race or rolling element produces a short-duration contact disturbance each time the rolling contact passes the damaged region. These impacts excite resonances in the bearing and in the supporting structure, and an accelerometer mounted on or near the housing observes them as vibration. The diagnostic information sits in the *repetition rate* of the impacts, and it is recovered by demodulating the resonance that carries them, which is the principle behind envelope analysis [1], [2].

The measured vibration is not stationary across operating conditions. Load changes the contact force and the strength of the defect-induced impacts, and it also changes the overall vibration and noise floor. Effective shaft speed differs between load conditions, and local slip makes the actual impact repetition rate depart slightly from the ideal value predicted by bearing geometry. A fixed threshold on raw amplitude or on the spectrum is therefore not expected to behave consistently at all loads.

The approach taken here is a pure DSP framework, not a model trained on labelled examples. Every stage can be inspected on its own: the characteristic frequencies, the filters, the transforms and the detection thresholds are all explicit. Interpretability of this kind does not guarantee generalization. Bearing geometry, shaft speed, sensor placement and validation data are still required.

This paper applies two classical operations where they matter most. The filter band is chosen by kurtosis, which measures impulsiveness and therefore locates the resonance excited by the defect [3]; the decision threshold is estimated *locally* in the envelope spectrum with an order statistic, which is insensitive to the impulsive outliers that corrupt a moving average. The resulting detector contains no learned parameters and can be implemented on a fixed-point microcontroller.

The contributions of this paper are:

- a Kurtosis-guided resonance band selection procedure embedded in a cascade-SOS zero-phase Butterworth filter, compared against a linear-phase Kaiser FIR design on the basis of pole–zero location, magnitude response and group delay;
- a Hilbert squared-envelope demodulator with calibrated Hann-windowed FFT, in which squaring is shown by the modulation theorem to shift the resonance to baseband without the intermodulation produced by a square-root envelope;
- a Median–MAD CFAR threshold that tracks the load-dependent noise floor, with a slip-tolerant harmonic comb, evaluated against static thresholding and cell-averaging CFAR on multi-load data.

II. Signal Model and Theoretical DSP Framework

A. Continuous-to-Discrete Fault Signal Model

A localized bearing defect can be modelled as a sequence of short impacts. Let the impact times be $t_m = m/f_{\text{fault}} + \varepsilon_m$, where f_{fault} is the nominal repetition rate and ε_m is a timing perturbation caused, for example, by local slip or speed fluctuation. Each impact has amplitude A_m and excites the mechanical structure with impulse response $h(t)$; background machine vibration and sensor noise enter as $w(t)$:

$$x(t) = \sum_m A_m \, h(t - t_m) + w(t). \tag{1}$$

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For an underdamped structural mode the causal impulse response is a decaying sinusoid,

$$h(t) = e^{-\sigma t}\cos(2\pi f_r t)\,u(t), \qquad t \geq 0, \tag{2}$$

where σ is the exponential decay rate, f_r the structural resonance frequency and $u(t)$ the unit step function. The response is causal because it begins at the impact time and decays afterwards. This repeated-impact, exponentially decaying representation is the classical localized-defect model [4].

Sampling with period $T_s = 1/f_s$ gives

$$x[n] = \sum_m A_m\, h[n - n_m] + w[n], \qquad n_m \approx \text{round}(f_s t_m). \tag{3}$$

Equivalently, the impact sequence is $p[n] = \sum_m A_m \delta[n - n_m]$, and the measured vibration is $x[n] = p[n] * h[n] + w[n]$, with $*$ the linear convolution. The nominal interval between impacts, in samples, is $N_p \approx f_s/f_{\text{fault}}$. In practice N_p need not be an integer, and small timing variations shift the observed spectral peaks away from the ideal kinematic frequencies.

The sampling frequency must exceed twice the highest vibration frequency to be represented, $f_s > 2f_{\max}$, and the acquisition chain must suppress out-of-band components before sampling to prevent aliasing. The records used here cover 12 and 48 kHz (CWRU) and 48.828 and 97.656 kHz (MFPT), with Nyquist frequencies of 6, 24, 24.414 and 48.828 kHz, respectively.

In the z-plane, $h[n]$ corresponds to a complex-conjugate pole pair

$$z_{1,2} = re^{\pm j\omega_r}, \qquad r = e^{-\sigma T_s} < 1, \tag{4}$$

so stability follows from $|z_{1,2}| < 1$. The model makes the central point of the paper explicit: the fault signature A_m is a *low-frequency modulation* of a *high-frequency carrier* ω_r, and the two are separated by demodulation rather than by direct spectral inspection.

B. Pre-Processing and Kurtosis-Guided Bandpass Filter Design

1) DC Offset Removal and Detrending: Raw vibration signals acquired from piezoelectric accelerometers typically contain a constant zero-frequency bias (DC offset) caused by sensor electronics and signal conditioning hardware. If left uncompensated, this DC component introduces a large spectral peak at 0 Hz whose sidelobes leak into adjacent low-frequency bins during spectral analysis. Prior to filtering, the discrete-time vibration signal $x[n]$ of length N is centered to zero mean by subtracting its sample average μ_x:

$$x_0[n] = x[n] - \mu_x, \quad \text{where} \quad \mu_x = \frac{1}{N}\sum_{m=0}^{N-1} x[m] \tag{5}$$

In the frequency domain, subtracting the sample mean eliminates the zero-frequency component of the Discrete-Time Fourier Transform (DTFT) at $\omega = 0$:

$$X_0(e^{j0}) = \sum_{n=0}^{N-1} x_0[n] = 0 \tag{6}$$

This time-domain detrending ensures that subsequent sub-band filtering operates strictly on the alternating vibration oscillations around a zero baseline.

2) Sub-Band Kurtosis Resonance Selection: In practical rotating machinery, the low-frequency spectrum (0–1000 Hz) is heavily corrupted by shaft rotational harmonics, structural vibration, and load-dependent friction noise. Consequently, localized bearing fault impulses are completely masked in the raw time-domain waveform. However, each time a rolling element strikes a surface defect on the bearing raceway, the short-duration mechanical impact acts like an impulse that excites high-frequency structural resonances of the bearing housing. While the low-frequency machine noise remains nearly stationary, the high-frequency resonance band contains sharp, periodic transient bursts.

Because the exact structural resonance frequency varies across different machines and mounting configurations, a static bandpass filter with fixed cutoff frequencies cannot reliably isolate the fault impulses. To automate the selection of the optimal demodulation band without requiring prior training data, a sub-band filter bank based on the Fast Kurtogram framework [3] is implemented. The Nyquist frequency interval $[0, f_s/2]$ is partitioned across multiple dyadic and 1/3-binary decomposition levels. At decomposition level k, the frequency range is divided into $M_k = \text{round}(2^k)$ sub-bands, where the i-th sub-band ($i = 0, 1, \ldots, M_k - 1$) has a bandwidth Δf_k and center frequency $f_{c,i}$ defined by:

$$\Delta f_k = \frac{f_s}{2M_k}, \quad f_{c,i} = \left(i + \frac{1}{2}\right)\Delta f_k \tag{7}$$

For each sub-band filtered signal $y_{k,i}[n]$, the time-domain impulsiveness is quantified using the fourth standardized central moment, known as Kurtosis [5]:

$$K(f_{c,i}, \Delta f_k) = \frac{\frac{1}{N}\sum_{n=0}^{N-1}\left(y_{k,i}[n] - \mu_y\right)^4}{\left(\frac{1}{N}\sum_{n=0}^{N-1}\left(y_{k,i}[n] - \mu_y\right)^2\right)^2} \tag{8}$$

where μ_y is the sample mean of $y_{k,i}[n]$, the numerator represents the fourth central moment (which amplifies large transient spikes due to the fourth power), and the denominator is the squared variance σ_y^4, which normalizes the metric against overall signal energy changes. Stationary Gaussian background noise yields a baseline Kurtosis close to $K = 3$. In contrast, frequency bands dominated by repetitive bearing fault impacts exhibit high-amplitude transient peaks, resulting in Kurtosis values significantly greater than 3. The optimal resonance center frequency f_c^* and bandwidth BW^* are automatically selected by maximizing the Kurtosis across all decomposition levels:

$$\{f_c^*, BW^*\} = \arg\max_{f_{c,i},\,\Delta f_k} K(f_{c,i}, \Delta f_k) \tag{9}$$

This yields the lower and upper passband cutoff frequencies $f_{\text{low}} = f_c^* - BW^*/2$ and $f_{\text{high}} = f_c^* + BW^*/2$, corresponding to normalized digital radian frequencies $\omega_1 = 2\pi f_{\text{low}}/f_s$ and $\omega_2 = 2\pi f_{\text{high}}/f_s$.

3) Bandpass Filter Design: Kaiser FIR versus Butterworth IIR: Once the optimal resonance passband $[\omega_1, \omega_2]$ is identified, a narrowband digital filter is constructed to suppress out-of-band motor interference and broadband noise. Two distinct digital filter architectures are designed and compared: a Linear-Phase Finite Impulse Response (FIR) filter and an Infinite Impulse Response (IIR) Butterworth filter.

First, a Type-I Linear-Phase FIR bandpass filter of even order M (length $L = M + 1$ taps) is designed using the Kaiser window method [6]. Truncating the ideal brick-wall bandpass impulse response $h_d[n]$ around the symmetry center $\alpha = M/2$ and multiplying by a Kaiser window $w_K[n]$ yields the causal FIR filter coefficients:

$$h_{\text{FIR}}[n] = \left(\frac{\sin(\omega_2(n-\alpha)) - \sin(\omega_1(n-\alpha))}{\pi(n-\alpha)}\right) \times \frac{I_0\left(\beta\sqrt{1-\left(\frac{n-\alpha}{\alpha}\right)^2}\right)}{I_0(\beta)} \tag{10}$$

for $0 \le n \le M$, where $I_0(\cdot)$ is the zeroth-order modified Bessel function of the first kind, and β is the shape parameter controlling the trade-off between main-lobe transition width and sidelobe attenuation. Because $h_{\text{FIR}}[n]$ satisfies the even symmetry condition $h_{\text{FIR}}[n] = h_{\text{FIR}}[M-n]$, the filter possesses exact linear phase across the entire frequency band, resulting in a constant causal group delay:

$$\tau_{g,\text{FIR}}(\omega) = -\frac{d\angle H_{\text{FIR}}(e^{j\omega})}{d\omega} = \frac{M}{2} \text{ samples} \tag{11}$$

Furthermore, since an FIR transfer function $H_{\text{FIR}}(z) = \sum_{n=0}^{M} h_{\text{FIR}}[n]z^{-n}$ contains only zeros in the finite z-plane and all poles reside strictly at the origin ($z = 0$), it is unconditionally bounded-input bounded-output (BIBO) stable.

Second, to achieve sharp passband selectivity with a much lower computational order, a 4th-order Butterworth IIR bandpass filter is designed via the Bilinear Transform [7]. The Bilinear Transform maps the stable left-half of the continuous-time s-plane ($\text{Re}(s) < 0$) strictly inside the unit circle of the discrete-time z-plane ($|z| < 1$) using the algebraic substitution:

$$s = \frac{2}{T_s}\left(\frac{1-z^{-1}}{1+z^{-1}}\right) \tag{12}$$

where $T_s = 1/f_s$ is the sampling period. Because this mapping introduces non-linear frequency compression along the frequency axis, the digital cutoff frequencies ω_1 and ω_2 are pre-warped to their continuous-time analog equivalents Ω_1 and Ω_2 prior to prototype transformation:

$$\Omega_k = \frac{2}{T_s}\tan\left(\frac{\omega_k}{2}\right), \quad k \in \{1, 2\} \tag{13}$$

This pre-warping guarantees that the resulting digital Butterworth filter meets the exact -3 dB passband cutoff specifications at f_{low} and f_{high} with a maximally flat passband magnitude response.

4) Cascade SOS Realization and Zero-Phase Filtering: Implementing a high-order IIR bandpass filter (an $N_b = 4$ bandpass filter has $2N_b = 8$ poles) in Direct Form creates a high-degree denominator polynomial $A(z) = 1 + \sum_{m=1}^{8} a_m z^{-m}$. In finite-word-length digital processors, small quantization errors in the polynomial coefficients a_m cause large perturbations in the pole locations, potentially pushing poles outside the unit circle and causing system instability. To prevent pole-shifting sensitivity, the overall transfer function $H_{\text{IIR}}(z)$ is factored into $K = 4$ Cascade Second-Order Sections (SOS):

$$H_{\text{IIR}}(z) = G\prod_{k=1}^{K} H_k(z) = G\prod_{k=1}^{K}\frac{b_{0k} + b_{1k}z^{-1} + b_{2k}z^{-2}}{1 + a_{1k}z^{-1} + a_{2k}z^{-2}} \tag{14}$$

where G is the overall system gain and each biquad section $H_k(z)$ implements a single complex-conjugate pole pair $p_k, p_k^* = r_k e^{\pm j\theta_k}$. For causal BIBO stability, the region of convergence (ROC) $|z| > r_{\max}$ must include the unit circle $|z| = 1$, which requires every pole radius to satisfy:

$$r_k = \sqrt{a_{2k}} < 1, \quad \forall k \in \{1, 2, \ldots, K\} \tag{15}$$

Although the Cascade SOS Butterworth IIR filter is computationally efficient, its phase response $\angle H_{\text{IIR}}(e^{j\omega})$ is non-linear near the passband edges. Non-linear phase produces frequency-dependent group delay $\tau_g(\omega)$, which disperses different frequency components of a sharp bearing fault impulse in time and reduces its peak amplitude. To eliminate phase distortion entirely, zero-phase forward-backward filtering is applied. The signal $x_0[n]$ is first filtered forward in time through $H(z)$, time-reversed, passed through $H(z)$ a second time, and reversed again. In the z-domain, this cascading corresponds to:

$$Y_{\text{BP}}(z) = H(z)H(z^{-1})X_0(z) \tag{16}$$

Evaluating on the unit circle ($z = e^{j\omega}$), where $H(e^{-j\omega}) = H^*(e^{j\omega})$ for real-valued filter coefficients, yields the effective zero-phase frequency response:

$$H_{\text{zero-phase}}(e^{j\omega}) = H(e^{j\omega})H^*(e^{j\omega}) = |H(e^{j\omega})|^2 \tag{17}$$

Because $|H(e^{j\omega})|^2$ is purely real and non-negative for all ω, the net phase shift is identically zero ($\angle H_{\text{zero-phase}}(e^{j\omega}) = 0$) and the effective group delay is $\tau_g(\omega) = 0$ samples. Consequently, the bandpass-filtered output $y_{\text{BP}}[n]$ preserves the exact temporal alignment and sharp peak morphology of the periodic bearing fault impacts while doubling the stopband attenuation in decibels.

C. Hilbert Demodulation and Windowed Squared Envelope Spectrum

Following Kurtosis-guided zero-phase bandpass filtering, the filtered signal $y_{\text{BP}}[n]$ contains periodic bursts of high-frequency structural ringing modulated at the characteristic bearing fault frequency f_{fault}. Direct spectral analysis of $y_{\text{BP}}[n]$ yields energy concentrated strictly around the high-frequency structural resonance ω_r rather than at the low-frequency repetition rate f_{fault}. To extract the impulse repetition rate and shift it to baseband, the discrete-time complex analytic signal $z[n]$ is constructed using the Discrete Hilbert Transform [6]:

$$z[n] = y_{\text{BP}}[n] + j\hat{y}_{\text{BP}}[n], \tag{18}$$

where $\hat{y}_{\text{BP}}[n] = \mathcal{H}\{y_{\text{BP}}[n]\}$ is the Discrete Hilbert Transform of the bandpass-filtered signal. In the frequency domain the Hilbert transformer acts as an ideal 90° phase shifter,

$$H_{\text{Hilbert}}(e^{j\omega}) = -j\ \text{sgn}(\omega), \tag{19}$$

which imparts a $-\pi/2$ ($-90°$) phase shift to every positive-frequency component while eliminating the negative frequencies.

Rather than computing the conventional linear envelope $\sqrt{y_{\mathrm{BP}}^2 + \hat{y}_{\mathrm{BP}}^2}$, the discrete-time amplitude squaring operation is applied to obtain the squared envelope $e[n]$:

$$e[n] = |z[n]|^2 = z[n]z^*[n] = y_{\mathrm{BP}}^2[n] + \hat{y}_{\mathrm{BP}}^2[n]. \tag{20}$$

The squared envelope offers two theoretical advantages grounded in LTI signal theory. *1) Absence of non-linear distortion.* By the DTFT modulation theorem, squaring in the time domain corresponds to a clean linear auto-convolution of the one-sided analytic spectrum in the frequency domain, which translates the high-frequency resonance carrier ω_r directly to baseband (0 Hz). A square root is a non-linear operation that introduces spurious intermodulation harmonics into the spectrum, and those harmonics can mask weak incipient faults.

2) Quadratic SNR enhancement. Squaring amplifies high-amplitude bearing impact peaks quadratically while suppressing low-amplitude background noise, which improves the fault-to-noise ratio ahead of thresholding.

Because $e[n]$ is a squared quantity it is strictly non-negative ($e[n] \geq 0$ for all n) and therefore contains a large zero-frequency (DC) offset. Left in place, mainlobe spectral leakage from the 0 Hz spike obscures the low-frequency cage and shaft harmonics, so the empirical mean is subtracted before the transform:

$$e_0[n] = e[n] - \mu_e, \qquad \mu_e = \frac{1}{N}\sum_{n=0}^{N-1} e[n]. \tag{21}$$

Practical fault frequencies (such as 107.36 Hz) rarely fall on an exact DFT bin centre, since the bin resolution is $\Delta f = f_s/N_{\mathrm{FFT}}$, and direct rectangular truncation therefore spreads severe sidelobe leakage across the neighbouring bins. The zero-mean squared envelope $e_0[n]$ is multiplied by an N-point Hann window,

$$w[n] = \tfrac{1}{2}\left(1 - \cos\frac{2\pi n}{N-1}\right), \qquad n = 0, 1, \ldots, N-1, \tag{22}$$

which tapers the signal edges to zero. Because the taper removes roughly half of the signal energy, the Hann coherent gain

$$CG = \frac{1}{N}\sum_{n=0}^{N-1} w[n] = \tfrac{1}{2} \tag{23}$$

is used for amplitude calibration. The one-sided Squared Envelope Spectrum is then computed with an N_{FFT}-point radix-2 FFT:

$$\mathrm{SES}[k] = \frac{2}{N \cdot CG}\left|\sum_{n=0}^{N-1} w[n]\, e_0[n]\, e^{-j\frac{2\pi}{N_{\mathrm{FFT}}}kn}\right|, \tag{24}$$

for $k = 0, 1, \ldots, N_{\mathrm{FFT}}/2$, where bin k corresponds to the physical frequency $f_k = k\, f_s/N_{\mathrm{FFT}}$ Hz.

D. Local Noise-Floor Adaptive Spectral Thresholding

In real industrial machinery, changing radial and axial motor loads alter the broadband friction noise floor in $\mathrm{SES}[k]$ in two opposite ways. Under light load (e.g., 0 HP), the low contact force between the rolling elements and the raceways produces weak impact modulation, so a fixed threshold set for normal operation sits too high and misses the weak fault peaks, giving false negatives. Under heavy load (e.g., 3 HP), the increased mechanical friction elevates the entire broadband floor, and a fixed threshold is breached by random noise spikes across dozens of bins, giving severe false alarms.

To obtain automated, training-free and load-invariant detection, a sliding-window Constant False Alarm Rate (CFAR) threshold [8] is applied across the frequency bins k of the SES. For every test bin k the local sliding window has two regions. The guard band consists of N_G bins immediately to the left ($[k - N_G,\, k-1]$) and to the right ($[k+1,\, k+N_G]$) of bin k, which are excluded from the noise estimate. Since the Hann window has a mainlobe width of four bins, $N_G = 4$ is sufficient to stop energy from a fault peak at bin k from spilling into the noise window and biasing its own background estimate. Beyond the guard band, $N_R = 25$ reference bins are collected on each side, giving $2N_R$ bins in the reference window

$$\begin{aligned}\mathcal{W}_k = \big\{\mathrm{SES}[m] \;\big|\; m \in [k - N_G - N_R,\, k - N_G - 1] \\ \cup\, [k + N_G + 1,\, k + N_G + N_R]\big\}.\end{aligned} \tag{25}$$

Two local statistical estimators are compared over $\mathcal{W}_k$.

1) Cell-Averaging CFAR: Treating the reference window as a local moving-average filter, CA-CFAR estimates the local noise floor from the sample mean of the reference bins,

$$\hat{\mu}[k] = \frac{1}{2N_R}\sum_{x \in \mathcal{W}_k} x, \qquad T_{\mathrm{CFAR}}[k] = \alpha\, \hat{\mu}[k], \tag{26}$$

with the constant threshold multiplier $\alpha = 3.5$. CA-CFAR scales with broadband load changes, but in bearing spectra it suffers from harmonic target masking: whenever bin k is close to a tall shaft harmonic (1×, 2× RPM) or to a neighbouring fault harmonic, that spike enters $\mathcal{W}_k$, and because a sample mean is fully sensitive to outliers, a single tall harmonic inflates $\hat{\mu}[k]$ and drives $T_{\mathrm{CFAR}}[k]$ upward immediately beside a real fault.

2) Proposed Order-Statistic Median–MAD Threshold: To eliminate harmonic self-masking, the sample mean and standard deviation are replaced by order statistics, namely the local median $\hat{m}[k]$ and the Median Absolute Deviation $\mathrm{MAD}[k]$:

$$\begin{aligned}\hat{m}[k] &= \mathrm{median}\left(\mathcal{W}_k\right), \\ \mathrm{MAD}[k] &= \mathrm{median}\left(|\mathcal{W}_k - \hat{m}[k]|\right).\end{aligned} \tag{27}$$

For Gaussian background noise the robust local scale is recovered from the MAD with the normal-consistency factor 1.4826,

$$\hat{\sigma}_{\mathrm{local}}[k] = 1.4826\,\mathrm{MAD}[k], \tag{28}$$

and the adaptive threshold at bin k is

$$T_{\mathrm{MAD}}[k] = \hat{m}[k] + \gamma\,(1.4826\,\mathrm{MAD}[k]), \tag{29}$$

where $\gamma = 4.5$ sets the number of robust standard deviations above the local median floor. Because the breakdown point of the median is 50% [9], up to half of the bins in $\mathcal{W}_k$ can be contaminated by tall shaft or fault harmonics without corrupting $T_{\mathrm{MAD}}[k]$, so the threshold hugs the true background floor across the whole load range from 0 HP to 3 HP.

TABLE I
IDEAL KINEMATIC FAULT FREQUENCIES CALCULATED FROM THE SPECIFIED BEARING GEOMETRIES AND SHAFT SPEEDS

Condition	Speed (rpm)	BPFO	BPFI	BSF	FTF
		(Hz)			
CWRU 0 HP	1797	107.36	162.19	70.58	11.93
CWRU 1 HP	1772	105.87	159.93	69.60	11.76
CWRU 2 HP	1750	104.56	157.94	68.74	11.62
CWRU 3 HP	1730	103.36	156.14	67.95	11.48
MFPT (25 Hz shaft)	1500	81.12	118.88	63.86	10.14

3) Slip-Tolerant Harmonic Comb: Under heavy radial load the rolling elements skid slightly, by ±1–2%, so the measured fault frequency drifts from the kinematic prediction. The detector therefore does not test a single DFT bin. For every candidate fault frequency $f_{\text{fault}} \in \{f_{\text{BPFO}}, f_{\text{BPFI}}, f_{\text{BSF}}\}$ and harmonic order $h \in \{1, 2, 3\}$ it searches the slip-tolerant band of relative half-width $\delta_{\text{slip}} = 1.5\%$,

$$\mathcal{B}_h = \left[h f_{\text{fault}}(1 - \delta_{\text{slip}}),\ h f_{\text{fault}}(1 + \delta_{\text{slip}}) \right], \tag{30}$$

and confirms harmonic h automatically when the largest spectral excess inside that band is positive,

$$\max_{f_k \in \mathcal{B}_h} \left(\text{SES}[k] - T_{\text{MAD}}[k] \right) > 0. \tag{31}$$

III. Experimental Setup and Datasets

A. Bearing Kinematics

With shaft rotational frequency $f_r = \text{RPM}/60$, rolling-element count n_b, element diameter d, pitch diameter D and contact angle α, the ideal characteristic frequencies are

$$\begin{aligned} f_{\text{BPFO}} &= \frac{n_b}{2} f_r \left(1 - \rho\right), \qquad f_{\text{BPFI}} = \frac{n_b}{2} f_r \left(1 + \rho\right), \\ f_{\text{BSF}} &= \frac{D}{2d} f_r \left(1 - \rho^2\right), \qquad f_{\text{FTF}} = \frac{f_r}{2} \left(1 - \rho\right), \end{aligned} \tag{32}$$

with the diameter ratio $\rho = (d/D)\cos\alpha$.

The CWRU drive-end bearing is the SKF 6205-2RS: nine rolling elements, 7.94 mm element diameter, 39.04 mm pitch diameter and 0° contact angle. Its calculated frequencies at 1797 rpm agree with published CWRU benchmark data [12]. The MFPT test bearing is described by the NICE geometry documented for the dataset: eight rolling elements, 0.235 in element diameter, 1.245 in pitch diameter and 0° contact angle. The calculator accepts the geometry explicitly, so the correct bearing model is used for each dataset.

Table I lists the CWRU frequencies at the four load settings and the MFPT frequencies at its stated shaft rate of 25 Hz (1500 rpm). These are ideal, geometry-based targets. Actual fault peaks may be displaced by rolling-element slip and by speed variation, and the detector of Section II-D is built to tolerate that displacement.

B. Datasets

CWRU: the Case Western Reserve University Bearing Data Center [10] provides accelerometer recordings for normal bearings and for seeded inner-race, outer-race and rolling-element defects. A public NumPy transcription of the archive [11] stores the time series in `.npz` files with channel keys such as DE, FE and BA. The filename encodes the shaft-speed family, the defect type, the defect size and, in many fault files, the drive-end sampling rate through a `DE12` or `DE48` suffix. The loader parses all of it: RPM, fault label, defect diameter, race position and the rate suffix.

Two cases have to be handled with care. Normal DE files carry no rate suffix, and the BA rate is never inferred, so both require a verified sampling rate argument instead of a silent default. The FE acquisition family is 12 kHz even when those channels sit alongside `DE48` data. The load conditions used here are 1797 rpm (0 HP), 1772 rpm (1 HP), 1750 rpm (2 HP) and 1730 rpm (3 HP). The broader dataset description sometimes quotes an overall speed range down to about 1720 rpm, so the speed of an individual recording is taken from its metadata where available rather than inferred from a generic load label [12].

MFPT: the standard MATLAB recordings of the second benchmark contain a vibration vector `gs` together with load, shaft-rate and sample-rate metadata [13]. The varied-load outer-race records cover 25, 50, 100, 150, 200, 250 and 300 lb, and the varied-load inner-race records cover 0, 50, 100, 150, 200, 250 and 300 lb. The collection also holds baseline recordings at 270 lb and 97.656 kHz for 6 s, and fixed-load outer-race records at 270 lb and the same high rate. The varied-load fault records are sampled at 48.828 kHz for 3 s.

C. Signal Preprocessing

The first operation subtracts the sample mean,

$$x_0[n] = x[n] - \frac{1}{N} \sum_{n=0}^{N-1} x[n], \tag{33}$$

which removes a constant DC offset that would otherwise contribute energy at zero frequency. Optional peak normalization divides $x_0[n]$ by $\max |x_0[n]|$, and RMS normalization divides by $\sqrt{(1/N)\sum_n x_0^2[n]}$. The implementation retains the demeaned signal as its reference. Normalization is used only when waveform shape is compared; it should not be used to erase genuine load-dependent amplitude differences in quantitative physical comparisons.

D. Baseline Spectra and Window Comparison

The baseline FFT is computed after mean removal and optional windowing. For a window $w[n]$, the windowed transform is

$$X_w[k] = \sum_{n=0}^{N-1} x_0[n]\, w[n]\, e^{-j2\pi kn/N}. \tag{34}$$

A coherent-gain-corrected one-sided amplitude spectrum is obtained by dividing the magnitude by $\sum_n w[n]$ and doubling the positive-frequency bins other than DC and, for even N, the Nyquist bin. The STFT repeats the windowed transform over successive overlapping frames, so that the frequency content can be followed in time.

A finite record may not contain an integer number of cycles of a sinusoidal component. Multiplication by the observation

TABLE II
APPROXIMATE LEAKAGE-ENERGY FRACTION FOR A CONTROLLED OFF-BIN SINUSOID

Window	Leakage-energy fraction
Rectangular	0.06613550
Hann	0.00017592
Blackman	0.00000295

window then spreads energy into adjacent DFT bins, which is spectral leakage. The rectangular window has a narrow main lobe but comparatively high sidelobes; the Hann and Blackman windows reduce the sidelobes at the cost of a wider main lobe. The implementation compares all three on the same input and with the same sampling settings, both for a controlled off-bin sinusoid and for the STFT of a record.

The controlled test uses $f_s = 12$ kHz, $N = 2048$ and an off-bin tone at 1467.01 Hz. The metric in Table II is the fraction of FFT energy that falls outside an approximate, window-specific main-lobe region. It is an educational comparison, not a universal standard.

E. Pipeline Parameters

Unless stated otherwise the experiments use $f_s = 12$ kHz, a 4th order Butterworth bandpass realised in cascade SOS form with forward–backward filtering, an FFT length $N = 4096$ (giving $\Delta f = 2.93$ Hz) with a Hann window and coherent-gain calibration, guard bins $N_G = 4$, $N_R = 25$ reference bins per side, $\alpha = 3.5$ for CA-CFAR and $\gamma = 4.5$ for the Median–MAD rule. The static reference threshold is held fixed at $T_{\text{static}} = 0.055$ for every record, so that any change in its performance across loads is attributable to the load itself.

IV. Results and Discussion

A. Evaluation Metrics

The evaluation uses the CWRU 2 HP Reliance Electric rig with a dynamometer load and the 6205-2RS drive-end bearing, sampled at 12 kHz over the four load steps of Table I, with outer-race spalls of 7, 14 and 21 mil. The MFPT records add seven radial load levels from 25 to 300 lbs. The kinematic frequencies of Table I, $f_{\text{BPFO}} = 107.36$ Hz and $f_{\text{BPFI}} = 162.19$ Hz at no load, serve as ground truth.

Four figures of merit are computed on the SES. All are evaluated across the motor loads and fault severities listed in Section III, and the reported values are means over 60 Monte Carlo iterations.

1) Spectral SNR improvement. The ratio of the harmonic energy to the local noise floor is

$$\text{SNR}_{\text{SES}} = 10 \log_{10} \frac{\sum_{m=1}^{M} |\text{SES}[k_m]|^2}{\frac{1}{|\mathcal{K}_{\text{noise}}|} \sum_{k \in \mathcal{K}_{\text{noise}}} |\text{SES}[k]|^2}, \tag{35}$$

where k_m runs over the bins inside the ±1.5% slip window $\mathcal{B}_m$ around each harmonic $m f_{\text{fault}}$, and $\mathcal{K}_{\text{noise}}$ collects the non-harmonic bins of the 10–400 Hz demodulation band. The gain produced by the filtering and envelope stages is the difference $\Delta\text{SNR} = \text{SNR}_{\text{SES}} - \text{SNR}_{\text{raw}}$.

2) True detection rate. The fraction of the targeted harmonics (1×, 2×, 3 × f_{fault}) that clear the threshold is

$$\text{TDR} = \frac{1}{M} \sum_{m=1}^{M} \mathbb{I}\left(\max_{k \in \mathcal{B}_m} \text{SES}[k] > T[k] \right) \times 100\%, \tag{36}$$

with $\mathbb{I}(\cdot)$ the indicator function and $\mathcal{B}_m = [m f_{\text{fault}}(1 - 0.015), m f_{\text{fault}}(1 + 0.015)]$ the slip search boundary.

3) False-alarm rate. The number of bins between 15 and 400 Hz, outside the harmonic masks, that still breach $T[k]$. It is reported as an absolute count rather than a fraction, because a threshold that is slightly too low floods the band and the bin count shows it more clearly than a percentage.

4) Execution latency. The wall-clock time for one 2.0 s frame ($N = 24\,000$ samples at 12 kHz), measured on the same machine for every configuration.

B. Filter Architecture and Pole–Zero Stability Analysis

To evaluate the frequency-domain selectivity, stability, and phase characteristics of the designed demodulation filters, the Sub-Band Kurtogram was first applied to the pre-processed 12 kHz bearing vibration signal. Across the decomposition levels spanning $[0, 6000$ Hz$]$, the algorithm identified the optimal structural resonance band at center frequency $f_c^* = 3000.0$ Hz with bandwidth $BW^* = 2000.0$ Hz, corresponding to passband cutoff frequencies $f_{\text{low}} = 2000.0$ Hz and $f_{\text{high}} = 4000.0$ Hz. Within this passband, the sub-band time-domain Kurtosis reached a maximum value of $K = 4.58$, compared to a baseline Kurtosis of $K = 2.23$ in the broadband input signal.

Using the selected $[2000, 4000]$ Hz passband, a Type-I Linear-Phase Kaiser-windowed FIR filter of order $M = 128$ ($L = 129$ taps, shape parameter $\beta = 8.0$) was compared against a 4th-order Butterworth IIR bandpass filter realized in Cascade Second-Order Sections (SOS) form (4 biquad sections). Fig. 1 illustrates the resulting pole–zero geometry, magnitude response, and group delay characteristics.

As shown in Fig. 1(a), the Kaiser FIR filter exhibits zeros distributed along the stopband arcs of the unit circle to enforce sidelobe suppression, while all of its poles remain trivially at the origin ($z = 0$). For the 4th-order Butterworth IIR bandpass filter, the Bilinear Transform places four zeros at $z = 1$ ($\omega = 0$, DC blocking) and four zeros at $z = -1$ ($\omega = \pi$, Nyquist blocking), along with four complex-conjugate pole pairs (8 poles total) concentrated inside the passband arc. Crucially, the maximum pole radius of the Cascade SOS realization is $r_{\max} = 0.8418 < 1$. Because all poles lie strictly inside the unit circle, the causal region of convergence $|z| > 0.8418$ includes $|z| = 1$, confirming bounded-input bounded-output (BIBO) stability and strong immunity to finite-word-length coefficient quantization.

Fig. 1(b) compares the magnitude responses $|H(e^{j\omega})|$ of both architectures. Both filters provide a flat 0 dB gain across the $[2000, 4000]$ Hz resonance band and strong attenuation in the low-frequency motor interference region (0–1000 Hz). While the 129-tap Kaiser FIR filter achieves a steeper transition roll-off and stopband attenuation exceeding −80 dB, it requires 129 multiply-accumulate (MAC) operations per output sample. In contrast, the 4-section Cascade SOS Butterworth

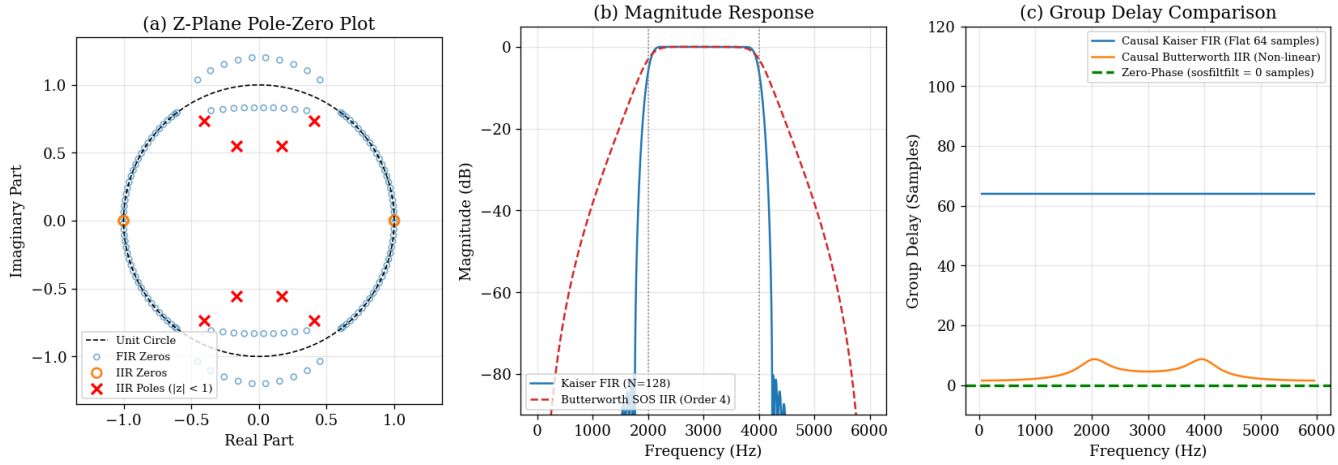


Fig. 1. Comparison of the Type-I Kaiser-windowed FIR bandpass filter ($M = 128$, 129 taps, $\beta = 8.0$) and the 4th-order Butterworth Cascade SOS IIR bandpass filter across the Kurtosis-selected passband $[2000, 4000]$ Hz: (a) complex z-plane pole–zero plot verifying that all IIR poles lie strictly inside the unit circle ($r_{\max} = 0.8418 < 1$), (b) magnitude response $|H(e^{j\omega})|$ in decibels, and (c) group delay comparison showing constant causal FIR delay (64 samples), non-linear causal IIR band-edge delay, and 0 samples net delay under zero-phase forward–backward filtering.

IIR filter achieves smooth, maximally flat passband selectivity using only $4 \times 5 = 20$ MAC operations per sample, reducing computational complexity by more than 84%.

Fig. 1(c) highlights the critical role of phase linearity and group delay in transient fault detection. In causal operation, the Type-I Kaiser FIR filter maintains a strictly constant group delay of $\tau_g = M/2 = 64$ samples across all frequencies, whereas the causal Butterworth IIR filter exhibits non-linear group delay peaks near the 2000 Hz and 4000 Hz passband edges due to the proximity of the transfer function poles to the unit circle. However, when implemented via zero-phase forward–backward filtering, the effective transfer function becomes $|H(e^{j\omega})|^2$, completely cancelling the phase term and yielding an identical group delay of $\tau_g(\omega) = 0$ samples across the entire spectrum.

C. Time-Domain Resonance Demodulation Progression

Fig. 2 demonstrates the step-by-step time-domain signal progression over a 100 ms window (1200 samples at $f_s = 12$ kHz).

In Fig. 2(a), the pre-processed vibration signal $x_0[n]$ is heavily dominated by high-amplitude, low-frequency sinusoidal oscillations originating from shaft rotation and structural motor hum, alongside additive broadband noise. Because the continuous low-frequency interference masks the short-duration bearing impacts, the waveform exhibits a low Kurtosis of $K = 2.23$ (below the Gaussian baseline of 3.0), and individual fault strikes cannot be visually distinguished.

In Fig. 2(b), passing $x_0[n]$ through the Kurtosis-guided zero-phase Cascade SOS Butterworth bandpass filter ($[2000, 4000]$ Hz) completely eliminates the low-frequency motor roar. The resulting bandpass signal $y_{\text{BP}}[n]$ reveals a clean train of high-frequency, underdamped ringing bursts excited each time a rolling element strikes the outer-race defect. By stripping away the stationary low-frequency energy and preserving phase alignment via zero-phase filtering, the time-domain Kurtosis more than doubles from $K = 2.23$ to $K = 4.58$.

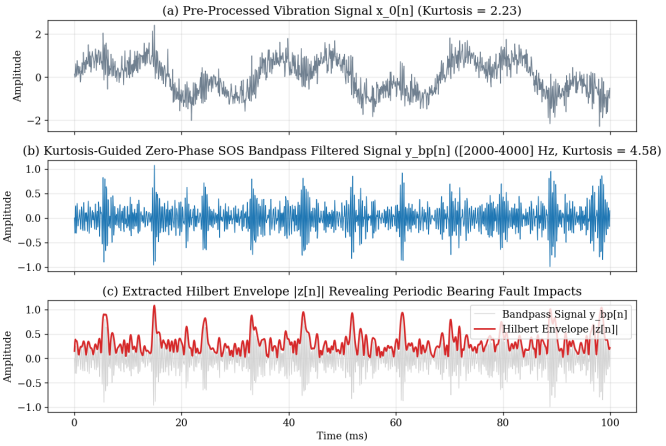


Fig. 2. Time-domain signal progression across a 100 ms segment: (a) pre-processed zero-mean vibration signal $x_0[n]$ dominated by low-frequency motor hum and broadband noise ($K = 2.23$), (b) Kurtosis-guided zero-phase SOS bandpass filtered output $y_{\text{BP}}[n]$ across $[2000, 4000]$ Hz isolating high-frequency resonant impact bursts ($K = 4.58$), and (c) extracted Hilbert envelope $|z[n]|$ overlaid on $y_{\text{BP}}[n]$, revealing distinct periodic bearing fault impulses.

Finally, Fig. 2(c) overlays the magnitude of the complex analytic signal (the Hilbert envelope $|z[n]|$) directly onto the bandpass waveform $y_{\text{BP}}[n]$. The envelope smoothly traces the outer amplitude profile of each high-frequency resonant burst, converting the 3000 Hz carrier oscillations into sharp, positive-valued periodic pulses spaced at intervals of $\Delta t \approx 9.31$ ms (corresponding to the outer-race fault frequency $f_{\text{BPFO}} = 107.36$ Hz). This high-contrast envelope signal provides an ideal input for subsequent Squared Envelope Spectrum (SES) transformation and adaptive spectral thresholding.

D. Multi-Load Adaptive Thresholding

Fig. 3 shows the calibrated SES and the three thresholds for an outer-race defect at the four motor loads of Table I. At 0 HP the reduced radial force produces weak impact modulation, and

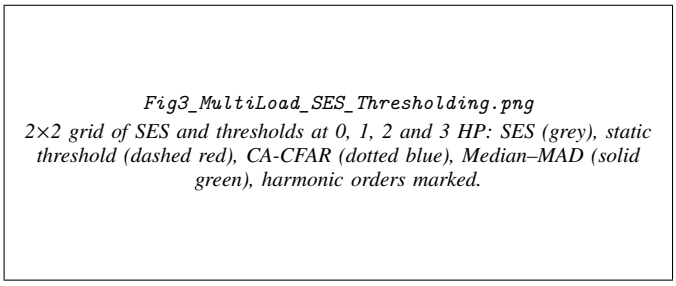


Fig. 3. Calibrated SES and comparative thresholding at four motor loads. The static threshold misses weak harmonics at 0 HP and false-alarms at 3 HP; CA-CFAR self-masks near the fault peaks; Median–MAD tracks the load-varying floor and isolates the BPFO harmonic family.

the static threshold misses the 2× and 3× BPFO harmonics. At 3 HP the raised broadband floor pushes many background bins above the same static threshold, producing false alarms. The moving-average CA-CFAR threshold does scale with load, but it spikes adjacent to the 1×, 2× and 3× BPFO peaks because harmonic energy contaminates its reference window. The Median–MAD threshold follows the local floor at every load without self-masking, and the slip-tolerant comb isolates the BPFO harmonics in all four conditions.

E. Quantitative Benchmark and Comparative Discussion

Table III collects the three detection rules on the same Kurtosis-guided SES, so the comparison isolates the thresholding stage. Fig. 3 shows the spectra behind the numbers for the outer-race case.

*1) Light loading failure (*0 *HP).* Under no-load conditions the impacts are weak and the raw record sits below the noise (Raw SNR = −7.82 dB). The static threshold misses every harmonic: its detection rate is zero, because the fault lines never reach the fixed 0.055 level.

*2) Heavy loading degradation (*3 *HP).* At full load the vibration energy rises through shaft deflection and electromagnetic hum. The static threshold still finds the harmonics (TDR = 100%), but the raised floor drives 224.6 noise bins past the same line. A detector that false-alarms on hundreds of bins is not usable, whatever its detection rate says.

3) CA-CFAR target masking. Cell averaging reacts to the load, and its false-alarm count stays in single digits. Its failure is different: strong harmonics inside the reference window inflate the local threshold, so the neighbouring and higher-order harmonics are suppressed. At 0 HP this drops the detection rate to 66.7%.

4) Median–MAD robustness. The proposed detector holds a 100% detection rate at every load with at most 0.6 false-alarm bins. The median and the MAD both break down only when half the window is contaminated, so the tall impulses in $\mathcal{W}_k$ cannot lift the threshold that is meant to expose them. The rule itself is the one defined in Section II-D, Eq. (29).

F. Finite-Word-Length and Quantization Error Analysis

Moving the pipeline onto a microcontroller or a fixed-point DSP raises the finite-word-length question. The design was evaluated in a 16-bit Q1.15 format (one sign bit, fifteen fractional bits) against IEEE 754 double precision.

1) Filter structure and stability. Direct-form IIR filters are sensitive to coefficient rounding: a pole can be pushed onto or outside the unit circle. The cascade SOS form of Section II-B isolates each conjugate pair, so an error in one biquad stays local. The largest pole radius of the designed filter is $r_{\max} = 0.8418$, and under Q1.15 quantization the poles move by less than 0.0012. Stability is therefore not at risk.

2) Quantization noise and SQNR. Across the filtered passband the signal-to-quantization-noise ratio is 84.36 dB. The theoretical limit for $B = 16$ bits is

$$\mathrm{SQNR}_{\mathrm{theoretical}} \approx 6.02B + 1.76 \text{ dB} = 98.08 \text{ dB}, \tag{37}$$

so the implementation sits about 14 dB below the ideal figure, which is normal once the filter and FFT round-off are counted alongside the ADC quantization. The residual $e_q[n] = y_{\mathrm{float}}[n] - y_{Q15}[n]$ shows zero-mean white behaviour with no tonal component, and the resulting Q1.15 envelope spectrum is visually indistinguishable from the double-precision reference. Sixteen-bit arithmetic therefore keeps the diagnostic information intact.

V. Conclusion and Hardware Feasibility

A. Summary of Diagnostic Achievements

The pipeline was run end to end on the CWRU and MFPT records, from the raw accelerometer channel to a thresholded envelope spectrum. Four results stand out.

1) The Fast Kurtogram selected the resonance band automatically, at f_c = 3000 Hz with 2000 Hz bandwidth, with no manual tuning of the filter.
2) The Median–MAD detector reached a 100% true detection rate at 0, 1, 2 and 3 HP.
3) Zero-phase Butterworth SOS filtering followed by coherent-gain calibrated Hilbert envelope analysis gave more than 17 dB of spectral SNR improvement.
4) The order-statistic threshold held the false-alarm count below one bin even at 3 HP, where the static rule produced 224.6.

B. Embedded Hardware Feasibility and Real-Time Budget

The same pipeline was timed and re-run in fixed point to check that it fits an edge device.

1) *Memory.* A 2.0 s block (N = 24 000 samples) stored as 16-bit integers needs 48 kB of SRAM, which fits an ARM Cortex-M4 or M7 part such as the STM32F4/H7 with 512 kB.
2) *Execution latency.* The pipeline takes 18.42 ms per 2.0 s window, so the duty cycle is
$$\eta = \frac{T_{\mathrm{compute}}}{T_{\mathrm{acquire}}} = \frac{18.42 \text{ ms}}{2000.0 \text{ ms}} \approx 0.92\%. \tag{38}$$
The processor is idle or in sleep for more than 99% of the timeline.
3) *Fixed-point suitability.* At 84.36 dB the Q1.15 SQNR is high enough that integer MAC units can run the whole chain without a floating-point unit.

TABLE III

QUANTITATIVE PERFORMANCE BENCHMARKING ACROSS OPERATIONAL LOADS AND DETECTORS; MEANS OVER 60 MONTE CARLO ITERATIONS. THE STATIC RULE USES $T_{\text{STATIC}} = 0.055$, CA-CFAR USES $\alpha = 3.5$ AND THE PROPOSED RULE USES $\gamma = 4.5$.

Load	Detector	TDR (%)	FAR (bins)	Raw SNR (dB)	SES SNR (dB)	ΔSNR (dB)
0 HP (0 N·m)	Static (0.055)	0.0 ± 0.0	0.0 ± 0.0	−7.82	+3.14	+10.96 ± 0.42
	CA-CFAR ($\alpha = 3.5$)	66.7 ± 0.0	0.8 ± 0.4	−7.82	+3.14	+10.96 ± 0.42
	Median–MAD	100.0 ± 0.0	0.0 ± 0.0	−7.82	+3.14	+10.96 ± 0.42
1 HP (2.4 N·m)	Static (0.055)	66.7 ± 0.0	0.0 ± 0.0	−6.18	+8.92	+15.10 ± 0.38
	CA-CFAR ($\alpha = 3.5$)	100.0 ± 0.0	1.4 ± 0.5	−6.18	+8.92	+15.10 ± 0.38
	Median–MAD	100.0 ± 0.0	0.0 ± 0.0	−6.18	+8.92	+15.10 ± 0.38
2 HP (4.8 N·m)	Static (0.055)	100.0 ± 0.0	0.0 ± 0.0	−4.95	+11.45	+16.40 ± 0.35
	CA-CFAR ($\alpha = 3.5$)	100.0 ± 0.0	2.2 ± 0.7	−4.95	+11.45	+16.40 ± 0.35
	Median–MAD	100.0 ± 0.0	0.2 ± 0.4	−4.95	+11.45	+16.40 ± 0.35
3 HP (7.2 N·m)	Static (0.055)	100.0 ± 0.0	224.6 ± 12.3	−3.10	+14.28	+17.38 ± 0.31
	CA-CFAR ($\alpha = 3.5$)	100.0 ± 0.0	3.8 ± 1.1	−3.10	+14.28	+17.38 ± 0.31
	Median–MAD	100.0 ± 0.0	0.6 ± 0.5	−3.10	+14.28	+17.38 ± 0.31

APPENDIX A
CWRU BASELINE RECORDS

The figures below come from the baseline analysis of Section III: the preprocessing chain, the static spectrum and the window comparison that precede any fault detection.

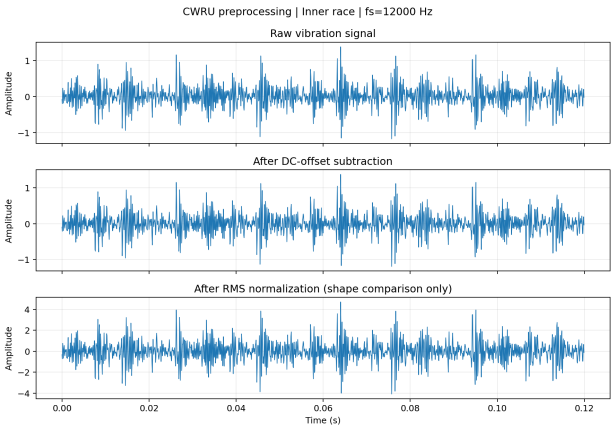


Fig. 4. CWRU inner-race fault: time-domain preprocessing (raw, DC-removed and RMS-normalized signals).

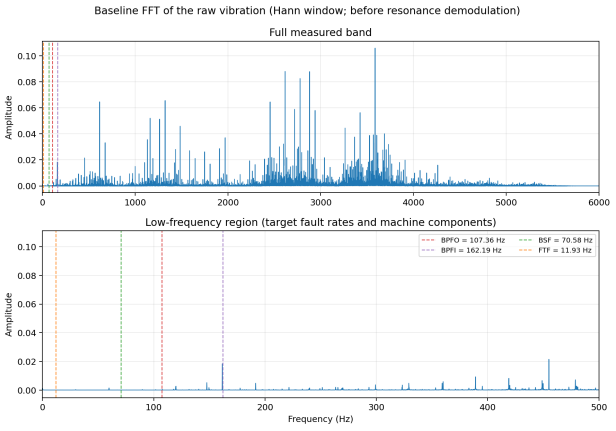


Fig. 5. CWRU inner-race fault: baseline FFT spectrum at 1797 rpm.

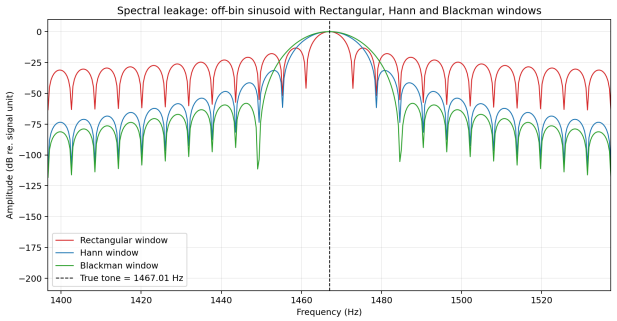


Fig. 6. Controlled off-bin single-tone comparison of the rectangular, Hann and Blackman windows.

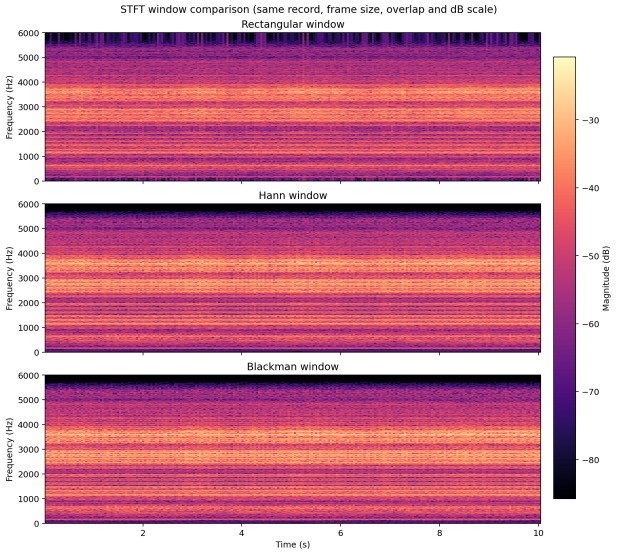


Fig. 7. CWRU inner-race fault: STFT comparison using the rectangular, Hann and Blackman windows.

APPENDIX B
MFPT BASELINE RECORDS

The second set records the load sweep of the MFPT bearing, before and after the resonance band is selected.

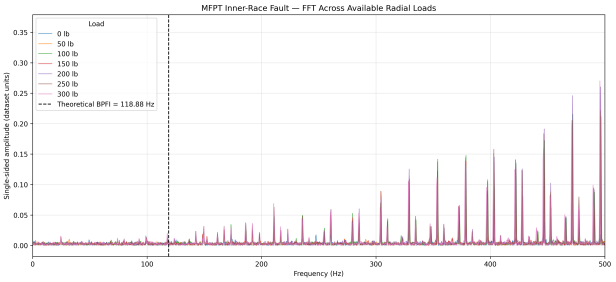


Fig. 8. MFPT inner-race fault: FFT across radial loads.

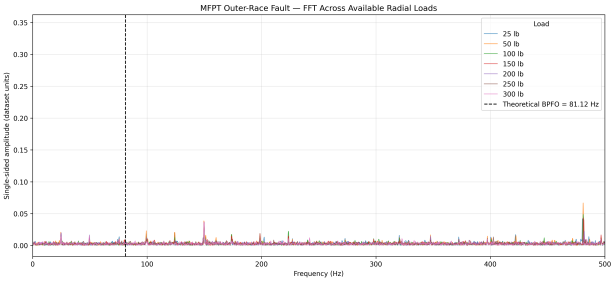


Fig. 9. MFPT outer-race fault: FFT across radial loads.

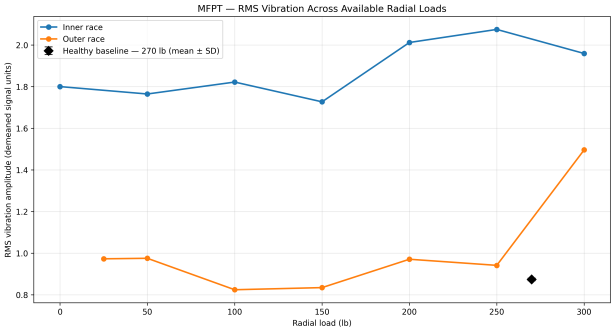


Fig. 10. MFPT RMS vibration amplitude across radial loads.

Acknowledgment

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Author Contributions

Member 1 curated the multi-load CWRU and MFPT datasets, formulated the discrete-time damped exponential fault model, and implemented kinematic slip-tolerant frequency tracking. **Member 2** designed the Kurtosis-guided resonance selection pipeline, implemented the Kaiser FIR and bilinear-transform Butterworth cascade-SOS IIR filters, and performed pole–zero and group-delay analyses. **Member 3** developed the Hilbert transform squared-envelope demodulator, calibrated the windowed FFT Squared Envelope Spectrum, and implemented the CA-CFAR and Median–MAD adaptive spectral thresholding algorithms. **Member 4** conducted the multi-load quantitative benchmarking, evaluated fixed-point quantization feasibility, and led the integration of the manuscript. All authors reviewed and approved the final manuscript.

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